

Homework 5

MEGN544/554 Robot Mechanics: Kinematics, Dynamics, and Control

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Problem 1. The ABB arm has the following DH table.

Link	a	d	α	θ
1	0	d_1	$\pi/2$	$\star$
2	a_2	0	0	$\star$
3	a_3	0	$-\pi/2$	$\star$
4	0	d_4	$\pi/2$	$\star$
5	0	0	$-\pi/2$	$\star$
6	0	d_6	0	$\star$

- (a) Draw the zero-angle configuration.
- (b) Solve the inverse kinematics (show all your work).
- (c) How many possible solutions exist?
- (d) Write a MATLAB code to demonstrate the efficacy of your script over 100 random test cases.